

1(a)(i)	capsid ; A capsomere A protein coat	1
1(a)(ii)	RNA / DNA ; (<i>actual is RNA</i>) I viral R dsRNA / dsDNA / mRNA / tRNA / rRNA	1
1(b)(i)	50 (nm) ; A 50 / 52.5 / 55 / 57.5 / 60 nm working showing use of correct formula, $A = I / M$; <i>length X–Y / ÷ magnification</i>	2
1(b)(ii)	ability to distinguish between two points ; AW	1
1(c)	malaria is transmitted by <i>Anopheles</i> (rather than <i>Aedes</i>) ; <i>plus any two from:</i> <i>max 1 if incorrect context stated for malarial transmission</i> both are transmitted by vectors ; both are transmitted by mosquitoes ; both, are blood-borne diseases / vectors feed on blood ;	3

Question	Answer	Marks
1(d)(i)	<i>max three if no correct ref. within response to, memory cells / immunological memory</i> A T- / B-cells, for, T- / B- lymphocytes <i>any three from:</i> 1 proteins are, <u>non-self</u> / <u>foreign</u>, <u>antigens</u> ; 2 proteins / antigens, stimulate a primary immune response ; 3 T-lymphocyte / B-lymphocyte, binding / recognition / clonal selection / activation / described in terms of receptor ; 4 clonal expansion / cell divides by mitosis to produce a clone / AW ; 5 formation of memory, (T- / B-) lymphocytes ; <i>mps 3, 4, 5 in context of primary immune response</i> 6,7 further detail ; e.g. (artificial) active immunity (T- / B-) memory cells, remain in, circulation / the body, for a long time or memory cells are long-lived (for secondary response) larger numbers of lymphocytes (specific to Zika virus) greater chance of encountering, Zika virus / pathogen / antigens secondary immune response is, stronger / faster (than primary immune response) faster production / higher concentration, of (specific) antibody (so) person does not have symptoms of disease when infected with (actual) Zika virus	4
1(d)(ii)	<i>any two from:</i> <i>ref. to herd immunity ;</i> A a large proportion of the population is <u>immune</u> (to Zika virus disease) ; <i>detail; e.g. context is less likely / less chance of / less risk of</i> people without immunity / unvaccinated people, less likely / unlikely, to be in contact with an infected person ora infected person unlikely to be in contact with a non-immune person <i>idea of increased chance of breaking the transmission cycle</i> AVP ; e.g. can achieve a high percentage cover if R_0 high, greater percentage of population need vaccinating ora	2